

Example 3.2-1 Consider the following LP with two variables:

$$\text{Maximize } z = 2x_1 + 3x_2$$

$$\text{subject to } 2x_1 + x_2 \leq 4 \rightarrow +s_1 \text{ (Slack)}$$

$$x_1 + 2x_2 \leq 5 \rightarrow +s_2 \text{ (")}$$

$$x_1, x_2 \geq 0$$

$$\max z = 2x_1 + 3x_2$$

$$\text{s.t. } 2x_1 + x_2 + s_1 = 4$$

$$x_1 + 2x_2 + s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

$$\begin{array}{ll} 2x_1 + x_2 + s_1 = 4 & \# \text{ of variables} = 4 \\ x_1 + 2x_2 + s_2 = 5 & \# \text{ of equations} = 2 \end{array}$$

$$n = 4, m = 2$$

$$n - m = 4 - 2 = 2 \text{ is assigned to zero}$$

$$C_4^2 = \frac{4!}{2! \cdot 2!} = \frac{4 \cdot 3}{2} = 6$$

- ① Nonbasic $x_1 = x_2 = 0$, Basic $s_1 = 4, s_2 = 5$; $z = 0$
- ② $x_1 = 0, s_1 = 0, x_2 = 4, s_2 = -3$ —
- ③ $x_1 = 0, s_2 = 0, x_2 = 5/2, s_1 = 3/2$; $z = 7.5$
- ④ $x_2 = 0, s_1 = 0, x_1 = 2, s_2 = 3$; $z = 4$
- ⑤ $x_2 = 0, s_2 = 0, x_1 = 5, s_1 = -6$ —
- ⑥ $s_1 = 0, s_2 = 0, x_1 = 1, x_2 = 2$; $z = 8$

⑥ $S_1=0$, $S_2=0$, $x_1=1$, $x_2=2$: $z=8$.

Optimum

$$\begin{array}{l} 2x_1 + x_2 + S_1 = 4 \\ x_1 + 2x_2 + S_2 = 5 \end{array} \rightarrow \begin{array}{r} 2x_1 + x_2 = 4 \\ -2 \mid x_1 + 2x_2 = 5 \\ \hline 2x_1 + x_2 = 4 \\ -2x_1 - 4x_2 = -10 \\ \hline -3x_2 = -6 \\ x_2 = 2 \\ x_2 = 2 \Rightarrow 2x_1 = 2 \\ x_1 = 1 \end{array}$$

- ① Model Formulation.
- ② Graphical Sol. Met. (2 variables)
- ③ Algebraic method (24)
- ④ Simplex Method (Adv, iterative)